Guruprasad
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**Preofessional Summary**

Analog Design Engineer with hands-on experience in LDO and power-management circuits in TI CMOS processes. Worked on Texas Instruments client projects involving block-level LDO design support and simulation, PVT and Monte- Carlo analysis, trimming, aging, and post-layout simulations. Experienced in silicon/IP-level characterization of non- volatile memories, VCO, HVGEN, and EEPROM IPs, including testbench development without data sheets. Strong back- ground in analog fundamentals, PMIC blocks, and Cadence Spectre-based design flow.

**Core skills**

**Analog Blocks:** LDO, Bandgap, Op-Amps, Comparators, POR, Voltage References

**PMIC Concepts:** Stability, PSRR, Load Transient, Compensation, Trimming

**Verification:** PVT, Monte-Carlo, Aging, Burn-in, Corner Analysis

**Simulation & Analysis:** AC, Transient, Noise, Stability analysis

**EDA Tools:** Cadence Virtuoso, Spectre, ADE

**Technology:** TI pdks

**Texas Instruments : Client Projects (via KarMic Design)**

1. **NMOS LDO Design:**
    - Bias circuit modification and block-level design support and verification of NMOS LDO.
    - PVT, Monte-Carlo, aging, and burn-in simulations
    - Line/load regulation, PSRR, stability, startup, and load transient analysis
    - Resistor trimming and post-layout simulation support

2. **Embedded Memory & NVM IP Characterization**
    - Worked on EEPROM, MTP, and NVM IPs across multiple TI technologies.
    - Developed testbenches for IDDQ, program/read currents, margin testing.
    - External IREF margin characterization across PVT and Monte-Carlo (up to 1000 samples per corner)
    - Post-burn and post-bake current analysis and data extraction.

3. **High-Voltage Generator & VCO IP Testing**
    - HVGENMINILV / HVGENMINIMVAL IP characterization.
    - Designed testbenches to measure IDDQ, frequency vs control voltage.
    - Verified VCO performance across temperature and process corners.
    - Supported performance improvement through circuit-level modifications.

**Analog and Mixed-Signal Design Projects (Training / Internal)**

1. **PMOS Capless LDO**
    - Designed and verified a PMOS cap-less LDO with focus on stability, quiescent current, and load regulation.
    - Conducted PVT and Monte-Carlo simulations to verify all the performance parameters and compared with the theoretical values.

2. **Bandgap Reference with trimming**
   - Designed and verified current mode sub 1V bandgap voltage reference using vertical PNP based PTAT, CTAT currents and high gain folded cascode operational amplifier.
   - Implemented a 5 bit resistor trimming circuit to tune the BGR across Process corners and mismatc

3. **Design and Implementation of Synchronous voltage mode CCM Buck Converter**
   - Designed a synchronous CCM buck converter comprising Type II compensated folded cascode amplifier, Com- parator, Level shifter, Non-overlapping single phase clock generator, Power switch driver, Power switches.
   - Verified each block and simulated Buck converter across PVT corners and measured performance parameters.

4. **Design and Implementation of 4 bit SAR ADC**
   - Designed a SAR ADC comprising Sample and Hold, Comparator, Successive Approximation Logic, Output buffer, Digital to Analog converter, Internal clock and control signal generator.
   - Verified its functional correctness across the PVT corners and measured essential performance parameters.

5. **Design and Implementation of Delta Sigma Modulator**
   - Designed a Delta Sigma Modulator comprising Fully differential folded cascode op-amp with common mode feedback, First order differential switched capacitor active integrator, Dynamic clocked comparator, One bit Digital to Analog converter, Non-overlapping two phase clock generator.
   - Verified each block and simulated the modulator across PVT corners and measured vital parameters.

6. **Design and verification of Digital Phase Locked Loop**
   - Designed a Digital PLL comprising Phase frequency Detector, Charge pump, Loop filter, Current starved 3 stage VCO, Divide by 2.
   - Verified across all PVT corners and ran Monte-Carlo simulation.

7. **Bandgap Reference in SKY130 â Tiny Tapeout Submission**
   - Set up an open-source analog design flow using Xschem, ngspice, Magic, Netgen, and the SkyWater SKY130 PDK.
   - Designed and verified a Bandgap Reference with a reference voltage of approximately 0.9 V through PVT corner and Monte-Carlo simulations.
   - Completed physical design, DRC, LVS, RC extraction, and post-layout verification.
   - Submitted the BGR design to Tiny Tapeout for fabrication using the SKY130 process; silicon is expected in December 2026.

8. **CMOS Comparator in SKY130 â Submission Ready**
   - Designed and verified a CMOS comparator using the SkyWater SKY130 PDK.
   - Completed schematic design, layout, DRC, LVS, RC extraction, and post-layout simulation.
   - Prepared the design as a submission-ready analog block following the Tiny Tapeout/SKY130 physical design and verification flow.